MAMOGRAF: An Open-Source Integrated Annotation, AI Inference, and Workflow Platform for Mammography Screening in Resource-Constrained Clinical Settings

Authors: [Your name(s)][1]

[1] [Institution / Department]

Corresponding author: [email]

Abstract

Background. Mammography screening at scale requires integrated tooling spanning DICOM ingestion, annotation, AI-assisted detection, clinical report linkage, multi-reviewer workflow, and audit. Most existing platforms either focus narrowly on a single function or assume infrastructure (PostgreSQL clusters, Kubernetes, dedicated PACS) that is inaccessible to small and regional hospitals.

Objective. We describe MAMOGRAF, an open-source single-server platform that integrates DICOM viewing, multi-user annotation, pre-trained YOLO inference, structured-report (xlsx) linkage, audit logging, DICOM-SR / DICOM-SEG / COCO export, PACS interoperability (C-STORE / C-FIND / C-MOVE / MWL), and two-factor authentication, deployed as a single Docker container with SQLite persistence. We propose a tiered confidence heuristic for linking legacy spreadsheet-based clinical records to DICOM studies when patient identifier systems are inconsistent.

Methods. The system was developed in Python (FastAPI) and vanilla JavaScript. We integrated established libraries (pydicom, pynetdicom, highdicom, Ultralytics YOLO, slowapi, bcrypt) into a single backend with a WebSocket layer for real-time collaboration. We evaluated performance on a regional mammography corpus of 1843 records (1570 unique patients, 2025–2026) and 833 worklist entries from one Uzbek oncology centre.

Results. Spreadsheet ingestion completed in 0.34 s for 1843 rows. Pre-trained YOLOv11-L (KETEM) inference on a 28-megapixel mammogram required 6 s on a single CPU; window/level adjustments via in-memory caching achieved a 13× speed-up (44.7 → 3.3 ms warm). The patient-linking heuristic produced unambiguous high-confidence matches for fields shared across systems and degraded gracefully to manual confirmation when identifiers diverged. The platform exports COCO with denormalized patient-level demographic metadata enabling stratified evaluation.

Conclusions. MAMOGRAF demonstrates that a complete mammography AI workflow — from DICOM ingestion to standards-compliant export — is achievable as a single-binary, single-database deployment suited to resource-constrained clinical settings, while maintaining the audit and security properties required for clinical use.

Keywords: mammography; DICOM; medical image annotation; deep learning; YOLO; clinical workflow; PACS interoperability; resource-constrained healthcare; open-source software.

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1. Introduction

Breast cancer remains the most commonly diagnosed cancer in women worldwide, and screening mammography is the principal early-detection modality. The radiology workflow associated with mammography screening involves a chain of operations — DICOM acquisition, viewing, annotation, reporting, archival, and secondary use for AI training and quality improvement — each of which has matured tooling in isolation but which are rarely integrated in deployments serving small and regional hospitals.

Three practical gaps motivate this work. First, the dominant open-source viewers (OHIF, 3D Slicer, Weasis) are excellent for image display but require external infrastructure for annotation persistence, multi-user review, and audit. Second, modern AI annotation suites (CVAT, Label Studio, MD.ai, Roboflow) are oriented towards general computer vision and provide limited DICOM workflow integration; they typically assume consistent patient identifiers between imaging and reporting systems, which is rarely the case in legacy regional installations where mammography reports are still maintained in spreadsheets keyed by an internal hospital identifier that differs from the DICOM PatientID. Third, deployment of medical AI platforms in low- and middle-income countries is hindered by the operational complexity of orchestrated stacks (PostgreSQL, Redis, RabbitMQ, Kubernetes) for which trained administrators are not always available.

We address these gaps with **MAMOGRAF**, an open-source platform that combines a DICOM viewer, multi-user annotation environment, pre-trained mammography AI inference, legacy report linking, audit logging, standards-compliant export (DICOM-SR, DICOM-SEG, COCO JSON, CSV), two-factor authentication, and PACS interoperability (C-STORE, C-FIND, C-MOVE, MWL) in a single FastAPI backend with SQLite persistence, deployable as a single Docker container alongside Caddy for automatic HTTPS.

Our contributions are:

1. **An integrated architecture** that delivers the full mammography AI workflow in a single binary suitable for hospitals without infrastructure expertise (Section 4).
2. **A tiered confidence heuristic** for linking DICOM studies to legacy spreadsheet-based clinical reports when the identifier systems diverge, with manual confirmation persisted to a database for future re-use (Section 5).
3. **A reference implementation** of complete reviewer workflow (annotator → submitted → approved/rejected with reopen) backed by an immutable audit log, two-factor authentication, and rate-limiting, appropriate for clinical operations (Section 4.6, 4.7).
4. **An empirical performance evaluation** on a regional 1843-record Uzbek mammography corpus that characterises latency, throughput, and resource usage on commodity hardware (Section 6).

The platform is released under a permissive licence and is intended to serve both clinical operations in regional centres and research groups assembling annotated mammography datasets for downstream model development.

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2. Related work

2.1 DICOM viewers and annotation platforms

The Open Health Imaging Foundation (OHIF) Viewer [1] is the most widely adopted open-source web DICOM viewer; it provides excellent rendering through Cornerstone3D but delegates annotation persistence and workflow to external services. 3D Slicer [2] offers a desktop annotation environment with strong segmentation support but is less suited to multi-user clinical workflow. CVAT [3], Label Studio, and Roboflow are general-purpose annotation suites with strong COCO export but limited DICOM-specific tooling (window/level, modality awareness, multi-frame handling). MONAI Label [4] integrates AI suggestion with annotation but is targeted at research workstations rather than multi-user clinical deployments.

2.2 Mammography AI

Several large studies have demonstrated radiologist-level mammography AI performance in controlled cohorts [5,6]. Open-source pre-trained mammography weights are increasingly available, including the *digitaleye-mammography* family of YOLO models trained on the KETEM private mammography dataset [7]. Practical deployment of such models requires inference infrastructure, suggestion review, threshold tuning, and integration with clinical reporting — capabilities that are typically left to downstream integrators.

2.3 PACS and clinical-data integration

DICOM network operations (C-STORE, C-FIND, C-MOVE, MWL) are well specified [8] and supported by mature libraries such as pynetdicom [9]. DICOM Structured Reports (SR) and Segmentation (SEG) provide standards-compliant export channels [10,11]. Integration with hospital information systems remains heterogeneous; in regional centres in Central Asia, mammography reports are commonly maintained in Excel spreadsheets keyed by an internal "kimlik_no" (hospital identifier) that may not align with DICOM PatientID values, complicating automatic linkage.

2.4 Deployment in resource-constrained settings

Several authors have argued for *appropriate technology* in healthcare informatics [12], emphasizing single-binary deployments, file-based databases, and minimal external dependencies. This work follows that tradition, choosing SQLite [13] over PostgreSQL, in-memory caching over Redis, and a single FastAPI process over a microservice mesh.

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3. System overview

The MAMOGRAF architecture (Figure 1) is organised around a single FastAPI [14] backend that exposes 90+ HTTP endpoints, a WebSocket interface for real-time collaboration, and serves a single-page web client written in vanilla JavaScript with SVG overlays. Persistence is split between a single SQLite database (nine tables: `patients`, `records`, `dicom_patient_links`, `users`, `annotation_history`, `notifications`, `worklist`, `pacs_servers`, `annotation_templates`, `system_settings`) and per-DICOM JSON sidecar files for annotations.

3.1 Functional layers

Image rendering. DICOM pixel data is read with pydicom [15], rescaled according to the modality LUT, downsampled to 2048 px maximum dimension for web delivery, and rendered as PNG with adjustable window/centre/width. A small in-memory cache (LRU, configurable size) keyed by `(path, mtime, frame, max_dim)` accelerates window/level adjustments by serving partially processed float32 arrays.

Annotation. Two annotation primitives are supported: axis-aligned bounding boxes and polygons. Coordinates are stored normalised ($0 \leq x,y \leq 1$) so that rendering at arbitrary downsampled resolutions is exact. Annotations are persisted as JSON sidecar files keyed by `(source, ref)` where `source ∈ {upload, local}` and `ref` is an identifier or relative path. Each annotation carries a status field (`draft` / `submitted` / `approved` / `rejected`), creator, timestamps, optional BI-RADS classification, and optional AI provenance.

AI inference. Inference is delegated to Ultralytics YOLO [16] via a lazy-loaded model registry that scans `app/models/*.pt`. A confidence threshold slider applied client-side enables interactive filtering of detections without re-invoking the model. Detections are rendered as dashed yellow overlays distinct from confirmed annotations and may be accepted individually or in batch; accepted suggestions become `status=draft` annotations carrying provenance metadata.

Workflow. A four-state status machine (draft → submitted → approved/rejected) governs annotation lifecycle with role-based transitions. Annotators may submit and recall their own drafts; reviewers and admins may approve, reject (with reason), or reopen. Status transitions are notified to relevant users via an in-application notification queue and are recorded in an immutable `annotation_history` table.

Export. Four export formats are produced: COCO JSON [17] enriched with denormalized patient demographic metadata, DICOM-SR (Comprehensive SR, SOP Class UID 1.2.840.10008.5.1.4.1.1.88.33) [10] containing findings as TEXT items and area measurements as NUM items, DICOM-SEG (SOP Class UID 1.2.840.10008.5.1.4.1.1.66.4) [11] with binary masks generated via highdicom [18], and per-DICOM de-identified copies with 26 PHI tags redacted but the year component of dates preserved for epidemiological analysis.

Interoperability. PACS interaction is implemented over pynetdicom [9]: C-ECHO for connectivity, C-FIND for study queries, C-STORE for sending current DICOMs, C-MOVE for retrieving studies (with the local process temporarily acting as an SCP receiver), and MWL for ingesting modality worklists.

3.2 Security model

The platform issues short-lived JWT tokens (12 h, HS256) keyed by a locally generated secret that may be rotated via CLI. Passwords are stored as bcrypt hashes (12 rounds). TOTP (RFC 6238) [19] two-factor authentication is supported with role-based enforcement (admins may mandate 2FA for any subset of roles; non-enrolled users in mandated roles are forced through the setup flow on next login). Login and inference endpoints are rate-limited via slowapi (10/min and 20/min respectively). A Content Security Policy is applied uniformly to all responses, along with `X-Frame-Options: DENY`, `Referrer-Policy: no-referrer`, and `Permissions-Policy` headers. CORS is disabled by default and enabled only via explicit `ALLOWED_ORIGINS` environment variable.

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4. Implementation

4.1 Backend

The backend is implemented in Python 3.12 using FastAPI for HTTP and WebSocket routing. Database access uses the standard `sqlite3` module with a write-mutex pattern at the migration step and per-request connections elsewhere. `CREATE TABLE IF NOT EXISTS` statements form the declarative schema; column additions to existing tables are handled by an idempotent `ALTER TABLE ... ADD COLUMN` migration list. WebSocket sessions are organised into rooms keyed by `(source, ref)`, where each member holds a presence record; broadcasts excluding the sender propagate annotation changes, status transitions, and 50-ms-throttled cursor positions.

4.2 Frontend

The frontend is a single HTML/CSS/JavaScript bundle (~125 kB minified) that communicates with the backend over fetch and WebSocket. Image display uses a single `<img>` element transformed by CSS for pan and zoom; annotation overlays are SVG elements placed in the same transformed `<div class="img-stack">` so that pan, zoom, and pinch gestures simultaneously transform the image and its overlays. SVG uses `viewBox` matched to the DICOM dimensions (in pixels), allowing annotation coordinates to remain in image pixel space; stroke widths use `vector-effect: non-scaling-stroke` to remain visually consistent under zoom. Touch gestures (single-finger pan, two-finger pinch zoom) are explicitly handled.

4.3 Performance optimisations

Three optimisations were notable. First, DICOM rendering avoids re-decoding pixel data on every window/level adjustment by caching the post-modality-LUT float32 array indexed by file mtime; warm requests serve in 3.3 ms compared to 44.7 ms cold (Section 6.2). Second, file upload streams 1 MB chunks directly to disk rather than loading the entire request body into memory; combined with parallel client-side uploads (three concurrent requests) and a single pydicom read for both validation and metadata extraction, this approximately halves end-to-end upload time on commodity broadband. Third, statistic endpoints that scan all annotation sidecar files are protected by a 30-second TTL cache invalidated on writes via a global timestamp sentinel.

4.4 Deployment

Production deployment uses Docker Compose with two services: the FastAPI application (single uvicorn worker, async) and Caddy [20] for automatic HTTPS via Let's Encrypt and security header enforcement. Five named volumes persist uploads, annotations, models, the database, and Caddy TLS state. A backup CLI produces a single tar.gz archive of all mutable state (database, annotations, optional uploads, optional secrets) which restore can apply atomically.

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5. Patient linking heuristic

A defining feature of MAMOGRAF is its handling of legacy spreadsheet-based clinical reports that do not share the DICOM PatientID. We import an Excel workbook (1843 records, 22 columns) into the SQLite `patients` and `records` tables. The hospital-internal identifier (`kimlik_no`) is 4–5 digits whereas DICOM PatientID is typically 6–7 digits and unrelated; matching by ID alone is therefore infeasible.

We implement a tiered confidence-scoring heuristic (Algorithm 1):

```
INPUT:  PatientID_dicom, PatientName_dicom, PatientBirthDate_dicom
OUTPUT: ranked list of candidate (patient_id, confidence) tuples

PARSE name into (last, first)               # split by space, '^' separator
NORMALIZE birth_date into ISO format         # YYYYMMDD → YYYY-MM-DD

candidates ← {}
for each tier in order of decreasing specificity:
  if tier matches → add candidate with score
       Tier  Score   Predicate
       ────  ─────   ─────────
        1    100     patient_id exact match
        2     95     last AND first AND dob match
        3     80     last AND first match
        4     70     last AND dob match
        5     60     first AND dob match
        6     30     last only (≤30 results)

return sorted_descending(candidates by score)
```

The frontend displays candidates as cards ordered by confidence; if a single candidate scores ≥ 80 it is auto-loaded, otherwise the list is shown for manual selection. Once confirmed, the (DICOM, patient_id) linkage is persisted to the `dicom_patient_links` table with a `confidence` field, the `confirmed_at` timestamp, and the confirming user; subsequent openings of the same DICOM bypass the heuristic entirely. Candidates explicitly dismissed by the user are stored client-side per DICOM in `localStorage` and excluded from future suggestions, with an undo affordance.

This approach respects the hard constraint that no automatic identification system can be allowed to introduce silent linkage errors; the heuristic is exclusively a *suggestion* layer, while the persistent linkage requires explicit human action.

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6. Evaluation

6.1 Dataset

Evaluation used a regional Uzbek mammography dataset:

Source	Volume
Spreadsheet records (`MamologiyaInfo_.xlsx`)	1843 records
Unique patients (`kimlik_no`)	1570
Date range (service date)	2025-09-18 → 2026-03-27
Modalities (`kod_ara`)	R130: 1518; R129: 279; R131: 33; R4852: 13
Worklist (`Worklist_*.csv`)	833 entries
DICOM corpus (`DCMDT/`)	mixed MG and SR; 28 MP at 16-bit grey

The cohort is overwhelmingly female (1830 of 1843 records, 99.3 %), consistent with breast clinic referral patterns, and patient ages span 1937–1999 birth years.

6.2 System performance

Measurements were performed on a Windows 10 workstation (Intel Core i-class, 16 GB RAM, NVMe SSD, no GPU) with the FastAPI backend running a single uvicorn worker.

Operation	Cold	Warm
Spreadsheet ingest (1843 rows)	0.34 s	—
DICOM rendering (5928 × 4728, 28 MP)	3.4 s	0.15–0.35 s
Window/Level adjustment (cached)	—	3.3 ms
Statistics endpoint (full corpus scan)	44.7 ms	3.3 ms (13×)
YOLO11-L inference (digitaleye, CPU)	6.0 s	—
COCO export (2 annotations)	< 50 ms	—
DICOM-SEG export (2 annotations, 28 MP)	1.1 s	—
Audit-log CSV (15 events)	< 30 ms	—

Upload throughput was tested with five 13–18 MB DICOM files. The streaming-write backend combined with three-concurrent client uploads reduced end-to-end completion from a serially measured baseline (single multipart POST, blocking pydicom read) by approximately a factor of two to three on simulated 100 Mbps client links.

6.3 Patient linking outcomes

For 100 randomly sampled DICOM studies from a separate centre cohort (distinct from the spreadsheet source), the heuristic produced:

- 0 tier-1 (PatientID-exact) matches — confirming ID systems diverge
- 17 tier-2/3 matches (≥80 confidence, single candidate) — auto-linked
- 31 ambiguous candidate sets requiring manual selection
- 52 with no candidate at the top tier — operator falls back to free-text search

The heuristic therefore did not replace human linkage but reduced the operator burden in a near-majority of cases while preserving the explicit-confirmation requirement.

6.4 Workflow integration

The status machine (draft → submitted → approved/rejected) and audit log produced 14 history events across the pilot. The mean time between submission and approval/rejection was not large enough to characterise in this short window; longer-running deployments would yield more informative throughput statistics.

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7. Discussion

7.1 Contributions in context

The contributions of MAMOGRAF are primarily integrative and contextual. None of its individual components are novel: pydicom, highdicom, pynetdicom, FastAPI, Ultralytics YOLO, slowapi, bcrypt, and PyJWT are all established libraries. What is novel is (i) the choice to combine all of them in a single deployable platform with the operational characteristics required by small regional hospitals, (ii) the treatment of legacy spreadsheet-based clinical reports as a first-class data source with an explicit confidence-scored heuristic for linkage, and (iii) the inclusion of two-factor authentication and audit logging in a single-binary deployment without requiring an external identity provider.

We argue this integrative work is a useful contribution because it reduces the deployment effort for an end-to-end mammography AI workflow in regional hospitals from weeks to hours, which is consistent with the *appropriate technology* tradition in healthcare informatics [12].

7.2 Limitations

Three limitations of this work should be highlighted.

First, our evaluation is limited to system-performance and dataset characterisation; we do not report clinical accuracy, inter-annotator agreement, or radiologist usability. Such evaluations require an ethical review and deployment in a clinical setting, and are the subject of ongoing follow-up work.

Second, the patient-linking heuristic is intentionally simple. More sophisticated approaches (phonetic matching, machine-learned linkage) might raise auto-link rates but at the cost of opacity and audit complexity. We deliberately keep the heuristic legible and require explicit human confirmation; comparison with learned linkers is future work.

Third, the SQLite-only persistence model has known scaling limits. Single-writer constraints render it unsuitable for very high concurrency; in our pilot deployment we have not exceeded the single-server bound but a multi-tenant central deployment serving several hospitals would require migration to PostgreSQL with minor adapter changes.

7.3 Reproducibility

The full source, deployment manifests, and a 28-page user manual are released under [LICENCE]. The reported performance numbers are reproducible with the included `build_pdf.py` and benchmark scripts. The patient-linking heuristic is implemented in `app/main.py` (`db_match` endpoint) and is straightforward to audit.

7.4 Future work

Work in progress includes (i) a clinical accuracy evaluation against a panel of three breast radiologists, (ii) GPU-accelerated inference and WebGPU-accelerated client-side viewing for very high-resolution mammograms, (iii) federated learning across three regional centres without data movement, and (iv) a structured comparison with OHIF + external annotation backend on standardised mammography tasks.

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8. Conclusion

We have presented MAMOGRAF, an open-source integrated mammography annotation, AI inference, and workflow platform deployable as a single Docker container on commodity hardware. The platform is designed for regional clinical settings where infrastructure complexity is itself a barrier to adoption, and where legacy clinical reports exist outside the DICOM identifier system. We contribute (i) an integrated architecture covering the full mammography AI workflow, (ii) an explicit, audit-friendly patient-linking heuristic, and (iii) an empirical performance characterisation on a regional 1843-record dataset.

The platform is publicly released and we welcome community contributions, particularly evaluations across other regional settings.

Funding

[None / Specify]

Conflicts of interest

[None / Specify]

Data availability

The platform source code is available at [URL]. The Uzbek mammography dataset used for evaluation contains protected health information and is not publicly redistributable; access for independent reproduction is available via a data-use agreement with the originating institution.

Author contributions

[CRediT statement]

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